

Semantic segmentation in remote sensing



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Outline

1. Introduction
2. From pixel to context
3. Markov and Bayesian models
4. Spectral-spatial and variational representations
5. Deep learning and foundation models
6. Open challenges



From image to thematic map

- ▶ Assign to **every pixel** a label from a predefined set of semantic classes [2] [3] [4]: a problem studied for decades under the names *supervised*, *per-pixel* and *contextual* classification
- ▶ Produces a **dense map**, to be distinguished from segmentation, which partitions the image into homogeneous regions *without* attaching any meaning to them
- ▶ In remote sensing: turning satellite observations into **land-cover and land-use maps**



(a) aerial image



(b) segmentation map

Zeebruges dataset, IEEE GRSS IADF Technical Committee; imagery and ground truth: Belgian Royal Military Academy and ONERA.

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- [3] P. H. Swain and S. M. Davis. *Remote Sensing: The Quantitative Approach*. McGraw-Hill, 1978
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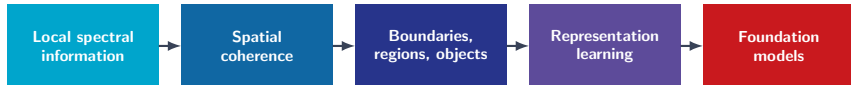


What sets remote sensing apart

- ▶ Heterogeneous **sensors**: optical, synthetic aperture radar (SAR), LiDAR.
- ▶ Variable spatial and spectral **resolutions**; **multitemporal** and **multimodal** acquisitions.
- ▶ **Complex geographical scenes** whose properties vary with region, season and acquisition conditions.
- ▶ The field is therefore not a simple adaptation of computer vision: it combines **statistical pattern recognition**, **signal processing**, **probabilistic modeling** and **deep learning**.

$$I : \Omega \longrightarrow \mathbb{R}^d \quad \text{then} \quad f : \Omega \longrightarrow \mathcal{C}$$

- ▶ Ω : the image domain; d : variables derived from spectral bands, SAR polarizations, LiDAR measurements, indices, DTMs, etc.
- ▶ $\mathcal{C}$: the finite set of semantic classes.
- ▶ The history of the field can be read as a **progressive integration** of spectral, spatial, contextual and then multimodal information.





The first approaches: per-pixel

- ▶ In the late 1970s: Landsat (optical) and SeaSat (radar) create the need for automatic land-cover maps [5] [6] [7]
- ▶ Each pixel is an observation vector $\mathbf{x} = (x_1, \dots, x_d)^T$: radiometric responses, SAR intensities, indices, ancillary data [2].

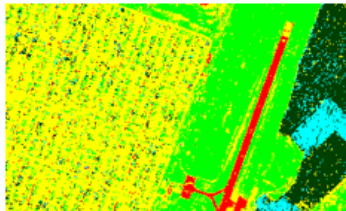
$$\hat{\omega} = \arg \max_{\omega_i} P(\omega_i | \mathbf{x})$$

- ▶ Under a multivariate Gaussian assumption: the **MAP classifier**, “maximum a posteriori”, the reference method for several decades [3] [4] [8]

- [2] R. O. Duda and P. E. Hart. *Pattern Classification and Scene Analysis*. Wiley, 1973
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Two limitations: dimensionality and independence

- **Curse of dimensionality** (Hughes phenomenon) [9]: more variables do not guarantee better performance when few samples are available.
- **Independence assumption on the observations**: each pixel is decided in isolation, without using its neighborhood, yet geographical objects are strongly spatially correlated: agricultural fields, forests, urban areas, road networks
- Consequence: **noisy maps**, all the more so as spatial resolution increases



Per-pixel classification: the salt-and-pepper effect [13]

[9] G. Hughes. "On the mean accuracy of statistical pattern recognizers". In: *IEEE Trans. Inf. Theory* (1968)

[13] G. Moser et al. "Land-cover mapping by Markov modeling of spatial-contextual information in very-high-resolution remote sensing images". In: *Proc. IEEE* (2013)



Discriminative and ensemble methods

- ▶ From the late 1990s: estimating **the decision boundaries directly**, rather than the distribution of the observations.
- ▶ **SVMs**: margin maximization; good generalization in high dimension [14] [15] [17] [18].
- ▶ **Kernels**: nonlinearity without losing convexity; many developments for geospatial observations [16].
- ▶ **Random forests**: a vote of decision trees learned on random subsets; robustness, little tuning, variable importance [19] [20].
- ▶ But these methods remain **essentially per-pixel** classifiers when they are applied to local observations.

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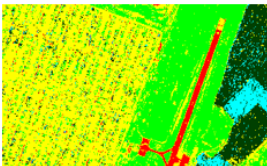
[20] P. O. Gislason et al. "Random forests for land cover classification". In: *Pattern Recognit. Lett.* (2006)



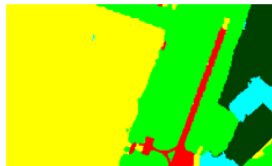
The contribution of spatial context



(a) IKONOS image



(b) per-pixel



(c) contextual

IKONOS image, 4 m, 3 bands, false-color composite; segmentation maps [13].

- In remote sensing, geographical objects exhibit **strong spatial consistency**: fields, forests, urban areas, road networks.
- Labels must be estimated **jointly**, not one pixel at a time.

[4] J. A. Richards. *Remote Sensing Digital Image Analysis*. Springer, 1986

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Markov random fields: the first formalization

- ▶ Labels become random variables defined on a **neighborhood graph** [10] [11] [22].
- ▶ Under the Markov assumptions and the equivalence with Gibbs distributions, MAP estimation becomes an **energy minimization**.

$$\hat{Y} = \arg \max_Y P(Y | X) \quad \Longleftrightarrow \quad \hat{Y} = \arg \min_Y U(Y | X)$$

- ▶ X : the observations; Y : the label map to be estimated.
- ▶ MRFs provide a general framework linking local data and spatial consistency.

[10] S. Geman and D. Geman. "Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images". In: *IEEE Trans. Pattern Anal. Mach. Intell.* (1984)

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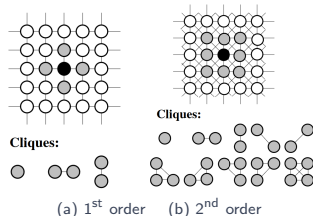
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Data term + regularization

$$U(Y | X) = U_d(Y | X) + \beta U_r(Y | X)$$

- ▶ U_d : the **data term**, derived from the likelihood.
- ▶ U_r : **spatial regularization**, promoting consistency between neighboring pixels.
- ▶ β : the trade-off between fidelity and regularity.
- ▶ 1st- or 2nd-order neighborhoods; clique potentials **penalize class changes** between **adjacent pixels** [22]



(a) 1st order (b) 2nd order
Pixel neighborhoods on a grid and the associated cliques [22].

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Optimization and uses of MRFs

- ▶ Finding the optimal configuration is a hard combinatorial optimization problem.
- ▶ Strategies: **ICM**, **simulated annealing** [23], the **Gibbs sampler** [10], **graph cuts** [24], loopy belief propagation [25].
- ▶ Applications: contextual classification, multisource segmentation, restoration, change detection, optical / radar / LiDAR fusion [12] [26] [27].
- ▶ Beyond these applications: a **general probabilistic formulation** of spatial regularization.

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From MRFs to CRFs: a bridge to deep learning

- ▶ Conditional random fields model $P(Y | X)$ directly rather than the joint distribution [28]: this makes **complex discriminative features** easier to use.
- ▶ **Fully connected CRFs**: better preserved object boundaries [29].
- ▶ Integration into deep networks: refinement modules, **differentiable layers** [30]; hierarchical Markov frameworks and learned CRFs [31] [32] [33] [34].
- ▶ **The idea that runs through the field**: a map must be consistent both with the local observations *and* with the spatial organization of the scene.

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Enriching the representation, not only the labels

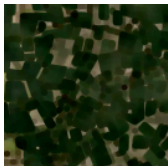
- ▶ Contextual models regularize the **labels**; spectral-spatial methods enrich the **features** before classification [6]
- ▶ As resolutions increase, radiometric signatures alone become insufficient.
- ▶ Texture, shape, size, spatial organization and neighborhood relations become **as discriminative as the spectral values themselves** [35].
- ▶ This rationale paves the way for the multiscale representations learned by deep networks [21].

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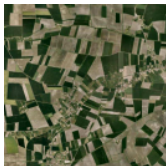
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Mathematical morphology and multiscale profiles



erosion



original RGB image



dilation

Erosion and dilation of the original RGB satellite image.

- ▶ Operators: erosion, dilation, openings, closings [36] [37].
- ▶ **Extended morphological profiles** and **attribute profiles** describe shape, size and contrast at several scales [38] [39] [40].

[36] J. Serra. *Image Analysis and Mathematical Morphology*. Academic Press, 1982

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Variational methods: regularizing the solution

- ▶ A complementary approach: integrating spatial consistency directly into the formulation of the problem.
- ▶ Segmentation is formulated as an **energy minimization** combining fidelity to the observations with spatial or geometric regularity [42].

$$E(u) = \text{data fidelity} + \lambda \text{ spatial/geometric regularity}$$

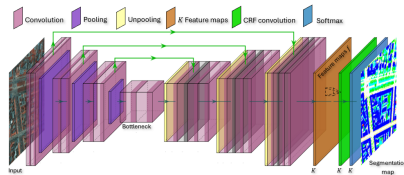
- ▶ The **Mumford–Shah** functional is one of the founding models [41].
- ▶ The principles of data/regularity trade-off, multiscale analysis and fusion will reappear in learned forms in deep learning.

[41] D. Mumford and J. Shah. "Optimal approximations by piecewise smooth functions and associated variational problems". In: *Commun. Pure Appl. Math.* (1989)

[42] C. Samson et al. "A variational model for image classification and restoration". In: *IEEE Trans. Pattern Anal. Mach. Intell.* (2000)

Going deep

- ▶ 2012, the turning point: **AlexNet** demonstrates a dramatic improvement in natural-image classification [43]
- ▶ The change: learning the representations and the decision function jointly, directly from the data.
- ▶ **FCNs**: dense, pixel-level prediction.
- ▶ **U-Net**: skip connections to restore spatial information.
- ▶ **DeepLab**: dilated convolutions and multiscale representations to enlarge the receptive field.

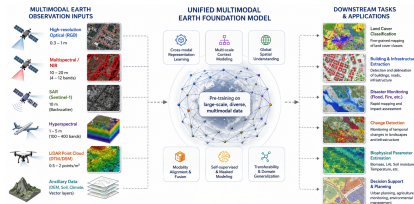


FCN architecture applied to Potsdam

- [43] A. Krizhevsky et al. "ImageNet classification with deep convolutional neural networks". In: *NeurIPS*. 2012
- [44] J. Long et al. "Fully convolutional networks for semantic segmentation". In: *IEEE/CVF CVPR*. 2015
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Transformers, self-supervision and foundation models

- ▶ CNNs remain limited by the need for annotations and by the modeling of long-range dependencies.
- ▶ **Attention** mechanisms and transformers better represent the interactions between distant regions [50] [51].
- ▶ **Self-supervision** exploits large volumes of unlabeled satellite images to learn transferable representations [52].
 - ▶ The goal: moving from a specialized model to **general, multimodal and transferable representations (geospatial foundation models)** [53].



Principle of a geospatial foundation model

- [50] A. Vaswani et al. "Attention is all you need". In: *NeurIPS*. 2017
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- [52] K. He et al. "Masked autoencoders are scalable vision learners". In: *IEEE/CVF CVPR*. 2022
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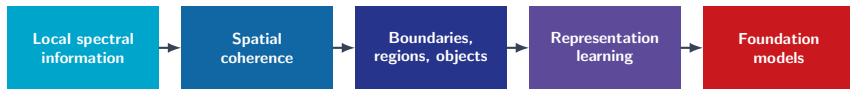
Open challenges

- ▶ **Generalization** across regions, seasons, sensors and resolutions.
- ▶ **Lack of annotations**: high cost of ground-truth data, class imbalance, noisy labels.
- ▶ **Multimodal fusion**: optical, SAR, LiDAR, time series and ancillary data.
- ▶ **Robustness**: distribution shift, clouds, noise, artifacts, rare events.
- ▶ **Interpretability**: understanding and controlling the decisions of increasingly large models.



Conclusion: a nonlinear history

- ▶ Not a simple trajectory from statistical classifiers to deep networks, but the **convergence of several families of methods**: statistical classification, kernel methods, ensemble methods, contextual models, Markov random fields, variational methods, stochastic geometry, deep learning, foundation models
- ▶ One unifying reading: the **progressive integration of increasing levels of context**



- ▶ One recurring principle: **data term + regularization**, from Markov random fields to the loss functions of deep networks

Thank you for your attention.





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